

Ryan Adidaru

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EDUCATION

University Of Indonesia

Bachelor of Engineering in Computer Engineering

Jakarta, Indonesia

2027

EXPERIENCE

AI Staff of Autonomous Marine Vehicle, University of Indonesia

Mar. 2024 – Present

University of Indonesia

Jakarta, Indonesia

- Architected and implemented autonomous navigation algorithms in ROS2, writing Python and C++ nodes to process sensor data and publish motor control commands for waypoint tracking.
- Currently developing modular ROS2 package integrating LiDAR, GPS/IMU, and camera subsystems, performing sensor calibration and real-time data fusion through Unity3d Simulation to enhance localization accuracy by 20%.
- Created and labeled custom image datasets for YOLOv4-tiny, YOLOv8, and YOLOv11 models using Roboflow; achieved up to 85% object detection accuracy in marine environments.
- Engineered OpenCV-based Python pipelines for color-based object segmentation, boosting recognition precision from 70% to 85%.
- Designed and deployed a Unity3D + ROS2 simulation platform in C# for ASV mission testing%.
- Streamlined team development workflows through GitHub branching strategies, code reviews, and CI pipelines, improving integration efficiency by 30%.

Hardware Intern of EXERCISE FTUI, University Of Indonesia

Oct. 2023 – Jan. 2024

University Of Indonesia

Jakarta, Indonesia

- Developed a platform for mushroom cultivation using IOT that produces temperature and humidity data
- Developed a software to maintain the mushroom automatically using linear regression's machine learning model that predicts optimal temperature and humidity for current settings which increases the productivity to 75 percent
- Further development of the software by utilizing AI Assistant-like models using character.ai unofficial api and google text-to-speech

HACKATHONS

DecentraLearn — ICP Hackathon 12 | *Rust, Internet Computer, Ollama, Qdrant*

Mar. 2025

- Co-developed a decentralized e-learning platform on Internet Computer Protocol, implementing Rust canisters for course, module, and lesson management with on-chain progress tracking.
- Integrated Ollama LLM canister (Llama 3.1) and Qdrant VectorDB to power AI-driven course recommendations and in-app chatbot tutoring.
- Designed modern frontend with React, TypeScript, TailwindCSS and shadcn/ui for responsive course exploration, authentication via ICP Internet Identity, and admin dashboards.
- Enabled secure, transparent student progress storage on-chain and real-time analytics visualizations using Recharts, enhancing learning insights for admins and instructors.

Nasa Space Apps Challenge 2024 | *Python, XGBoost, Pandas, Scikit-Learn*

Sep. 2024

- Developed end-to-end predictive pipeline using NASA climate datasets (rainfall, soil moisture, solar radiation, temperature) and local market price data for fruits and vegetables.
- Engineered geospatial and temporal features; trained XGBoost regression models achieving competitive RMSE for environmental and price forecasts.
- Visualized model performance and data trends with Matplotlib and Seaborn; collaborated within a 4-member team to deliver a working prototype in 48 hours.

PROJECTS

- RoR — RewrittenOnRust** | *Rust, Machine Learning, Ollama, VectorDB (Qdrant)* Mar. 2025 – Present
- Initiated and currently developing open-source project in Rust that applied popular LLMs stack, such as RAG using Qdrant as VectorDB and MCP, also i'm using ollama as backend endpoint for LLM model and Embedding model
- asvsim** | *Python, C++, C#, Unity* Dec. 2024 – Present
- Forked and contributed to open-source ASV robotic simulation and currently still developing autonomous system to integrate in ROS2
- yolo.cpp** | *CUDA, C, C++, Neural Networks* Jul. 2024 – Dec.2024
- Developed an open-source project for a vision model from scratch ranging from C++ with only using omp.h as an attempt to maximize the CPUs to CUDA for GPU Optimization
- amadeus** | *NLP, Groq, Llama 3.1* Jul. 2024 – Present
- Developed a personal assistant with langchain that can retain memory from previous conversation with the users with combined APIs from Groq for the LLMS and Deepgram for text-to-speech.
- mitsuki** | *Python, Streamlit, Llama 2.70b, NLP* Jan. 2024 – Feb. 2024
- Trained Llama 2 with custom datasets, to achieve desired personallity for the chatbot and deploy the chatbot with streamlit
 - Made and used my custom dataset to train the Llama 2
 - Implemented GitHub OAuth to get data from user's repositories
- Trained YOLOv4-Tiny** | *Computer Vision, YOLOv4-tiny, OpenLabelling, Google Colab* Jan. 2024 – Jan. 2024
- Developed a customized YOLOv4-tiny model using my datasets that i created using AyamBawang-dataset which consists of 300 labeled images of indomie
- emotionDetector** | *Computer Vision, Machine Learning, Google Colab* Nov. 2023 – Nov. 2023
- Developed a customized machine learning models that can recognizes human emotions, starting from happiness to fear, results of Independent Research that i conducted
- Digital Mouse With Hand Recognition** | *Computer Vision, Machine Learning, Google Colab* Oct. 2023 – Oct. 2023
- Developed a customized machine learning models that can recognizes Hand gestures that can be converted to digital mouse with certain gestures, pinching for left click, etc. results of Independent Research
- WaifuAI** | *NLP, Machine Learning, Google Colab* Oct. 2023 – Oct. 2023
- Utilized CharacterAI unofficial API to build AI Assistant that can act like Hu-Tao from Genshin Impact and can speak with help from text-to-speech from google, results of Independent Research
- SEA Countries Education Analysis** | *Visualization, Kaggle, Machine Learning, Google Colab* Nov. 2023 – Nov. 2023
- Visualized data of SEA Countries in term of education with custom datasets from Kaggle, results of Independent Research

CERTIFICATES

IBM Certified Applied AI Professional
Google Advanced Data Analytics
Google Foundations of Data Science
Google Get Started With Python
Google Advanced Data Analytics Capstone
Google Go Beyond the Numbers: Translate Data into Insights
Google Regression Analysis: Simplify Complex Data Relationships
Google The Nuts and Bolts of Machine Learning
Google The Power of Statistics
Sololearn Python Intermediate
Udemy Web Developer Bootcamp

TECHNICAL SKILLS

Languages: Rust, Python, C/C++, JavaScript, HTML/CSS, SQL, CUDA
Developer Tools: Git, Jupyter Notebooks, Google Colab, VS Code, Visual Studio, Kaggle, Ubuntu 22.04, neovim, vim
Libraries: pandas, NumPy, Matplotlib, OpenCV, ROS, ROS2, YOLO Architecture